

- There are 3 Questions for 15 marks.
- Write your answers neatly, legibly and logically. Keep your proofs thorough. Your proofs must follow from the basics discussed in class.
- You do not need any calculators.
- Good luck!

1. Consider the sequence of functions ($f_n^k : [0, 1] \mapsto \mathbb{R}$)

$$f_n^k(x) = \begin{cases} 2^{n/2} & \text{for } k-1 \leq 2^n x \leq k-0.5 \\ -2^{n/2} & \text{for } k-0.5 \leq 2^n x \leq k \\ 0 & \text{otherwise.} \end{cases}$$

Here $1 \leq k \leq 2^n$ and $n \geq 0$ are non-negative integers.

(a) [3] Consider an equation of the form

$$a_{n_1}^{k_1} f_{n_1}^{k_1}(x) + a_{n_2}^{k_2} f_{n_2}^{k_2}(x) + \dots + a_{n_r}^{k_r} f_{n_r}^{k_r}(x) = 0,$$

where (n_i, k_i) are distinct pairs, and $a_{n_i}^{k_i}$ are constants/real numbers. Given the pairs (n_i, k_i) , explain how to find the constants $a_{n_i}^{k_i}$.

(b) [3] How would you characterize the span of the following set of functions ?

$$\bigcup_{n \leq 2} \bigcup_{k=1}^{2^n} \{f_n^k(x)\}$$

2. Let $\mathcal{S}$ be the set of real sequences (a_n) that satisfy the following recurrence

$$a_n = a_{n-1} + a_{n-2}, n \geq 2.$$

(a) [3] Show that $\mathcal{S}$ is a vector space (with the standard addition and scaling operations) over $\mathbb{R}$.

(b) [3] What is the dimension of this vector space ?

3. [3] Is $\mathbb{C}$ a vector space over $\mathbb{R}$? If so, are 1 and i linearly independent ? What is the dimension of this vector space ?